Preliminary Assessment of CICERO Radio Occultation Performance by Comparing with COSMIC I Data

Hyeyeon Chang and Jiyun Lee
Korea Advanced Institute of Science and Technology

Yang Wang, Brian Breitsch and Y., Jade Morton
University of Colorado Boulder

BIOGRAPHY (IES)

Hyeyeon Chang is a doctoral student in Aerospace Engineering at Korea Advanced Institute of Science and Technology (KAIST). She received the B.S. degree from KAIST. She worked on assessment of ionospheric spatial decorrelation for Ground-based Augmentation System (GBAS) operation. Her current research interests are in GNSS radio occultation for atmosphere/ionosphere modeling.

Dr. Jiyun Lee is an Associate Professor in the school of Mechanical Aerospace and Systems Engineering at KAIST. As part of her professional experience, she worked as a Consulting Professor with Stanford University, a Principal Systems Engineer with Tetra Tech AMT, and a Senior GPS Systems Engineer with SiRF Technology, Inc. Her specialties lie in the area of integrity architecture, GNSS augmentation systems, and atmospheric and space weather monitoring. She received her Ph.D from Stanford University in aeronautics and astronautics. She served ION as Asia-Pacific technical representative (2017-2018), ION GNSS+ track chair, ION ITM program/general chair, and an associate editor of NAVIGATION.

Yang Wang is a doctoral student in aerospace engineering sciences at the University of Colorado Boulder. He received a B.E. from Beihang University in 2013 and an M. E. Degree from Nanyang Technological University in 2016. His research focuses on GNSS radio occultation and reflectometry signal processing and applications.

Brain Breitsch is a graduate student at the University of Colorado, Boulder. He received undergraduate degrees in Electrical Engineering and Mathematics from Miami University in Ohio, and a Masters in Electrical Engineering from Colorado State University. His research interests include estimation and error mitigation techniques applied to GNSS remote-sensing applications.

Dr. Jade Morton is a professor at the Aerospace Engineering Sciences Department at the University of Colorado (CU) Boulder. Prior to joining CU in 2017, she was an electrical engineering professor at Colorado State University (2014-2017) and at Miami University (2000-2014). Dr. Morton's research interests lie at the intersection of satellite navigation technologies and remote sensing of the Earth's space environment, atmosphere, and surface. She received a PhD in Electrical Engineering (EE) from Penn State and was a postdoctoral research fellow at the University of Michigan. Dr. Morton is an Institute of Navigation (ION) Thurlow and Burka award winner, and an IEEE Kershner award winner and Distinguished Lecturer. She is the president of ION, a fellow of IEEE, the ION, and the Royal Institute of Navigation.

ABSTRACT

Community Initiative for Cellular Earth Remote Observation (CICERO) is a planned constellation of low-earth-orbiting 6U cubeSats for performing GNSS radio occultation (RO) of Earth's atmosphere and surface. The goal of CICERO is to provide the first and high-quality commercial radio occultation data from space at low costs while enhancing weather and climate forecasting capabilities. Before CICERO data are used for reliable weather forecasting, the assessment of its performance is necessary. This study shows the performance of CICERO by comparing it with that of COSMIC I. The performance analysis was carried out with respect to geographic distribution, altitude distribution, and refractivity error. The results show that CICERO can acquire global coverage of data at low altitudes as well as COSMIC I, which is important for climate research. In addition, the analysis of refractivity error and its impact on temperature demonstrates that the miniature version of the GNSS RO receiver could satisfy certain accuracy requirements of the GNSS RO measurements. Thus, the cubeSat constellation CICERO can provide radio occultation measurements comparable to those of the COSMIC I mission. The study demonstrates the capabilities of nanosat-based LEO cubeSats to improve data obtainability and accuracy for weather forecasting.

1.0 INTRODUCTION

With the development of GNSS Radio Occultation (GNSS RO), there has been a significant improvement in numerical weather prediction (NWP) models for operational forecasts [1]. GNSS RO is a remote sensing technique that derives weather parameters using GNSS signals received by low-earth-orbit (LEO) satellites. As the GNSS signal passes through the atmosphere, the signal is delayed and its ray is bent. By measuring the signal delay and bending, GNSS RO provides fundamental measurements of key parameters including refractivity, temperature, pressure, and water vapor, which are required in climate modeling and weather forecasting [2]. One of the major contributions of GNSS RO to recent climate research is the enhanced availability of observations with a global coverage. While the profiles from other atmospheric observation instruments such as radiosonde are limited to local land areas, GNSS RO profiles can be obtained from global land and ocean areas. Furthermore, GNSS RO has strengths of high vertical resolution, and all-weather capabilities [3]. With these advantages, GNSS RO could improve data obtainability which brings a significant benefit to climate research.

As the demand for better numerical weather prediction models increases, there have been many efforts to increase the number of GNSS RO observations [4-6]. The amount of observation data is directly related to the number of daily occultation profiles and spatial observation samples. A larger number of GNSS RO observations could be obtained with the use of a larger number of LEO satellites equipped with GNSS RO receivers [7]. The development times and cost are major factors that hinder the realization of a large number of LEO satellites for RO applications.

The Community Initiative for Cellular Earth Remote Observation (CICERO) is a recent constellation deployed by GeoOptics to demonstrate the feasibility of obtaining quality RO data using low cost cubeSats. The business concept of the CICERO is provision of reliable, regular, low cost data with the augmentation and continuation of the Constellation Observing System for Meteorology Ionosphere and Climate (COSMIC) I and II [8]. Unlike early high-end receivers in COSMIC (e.g., BlackJack, TriG) [9], the CICERO cubeSats carry relatively low-cost, low-power, and low mass receivers called CIONs [10]. Developed by JPL, the CION is a miniature GNSS receiver, hosting a 2×3 patch antenna assembly. CION fits within a 30×10×6-cm envelope and has a mass of 1.2 kg [11, 12]. The miniaturization of the GNSS RO receiver allowed a CICERO payload to be fit into a 6U cubeSat. Taking advantage of CICERO’s characteristics, it is possible to launch many small satellites to acquire massive occultation data at low costs.

Before CICERO data are used for reliable weather forecasting, the assessment of its performance is necessary. In this study, we conducted the following CICERO data analysis for its performance assessment: 1) geographic distribution of the RO measurements, 2) altitude distribution (penetration altitude) of the RO measurements, and 3) refractivity error with respect to radiosonde profiles. The geographic distribution of RO occultation profiles is an important subject because it affects the spatial resolution of the measurement and global data acquisition capability. The altitudinal analysis, especially the low altitude penetration capability, is also necessary for measuring the key weather parameters in the lower troposphere. The analysis of refractivity error is essential as the refractivity is the primary measurements from which atmospheric parameters such as temperature, density, and water vapor content can be derived [2, 13-15]. Because COSMIC I is an established constellation that defines the state-of-the-art RO measurements, COSMIC I measurements collected within the same period of CICERO measurements are used for comparison.

In this paper, we show that the cubeSat constellation CICERO can provide radio occultation measurements comparable to the COSMIC I mission. This paper is organized as follows: Section 2 describes the data used for the performance assessment. The methodology for performance analysis is presented in Section 3. Section 4 shows the results of the performance assessment with discussions. Finally, the conclusions of this study and plans for future work are included in Section 5.

2.0. DATA

CICERO data collected in June-July 2019 are used for the analysis. COSMIC I data and collocated radiosonde profiles for comparison are obtained from the COSMIC Data Analysis and Archive Center (CDDAC). There were total of 12 days for which both COSMIC I/radiosonde and CICERO profiles were available in June-July 2019. The details of the data are described in the following subsections.

2.1. CICERO data

The CICERO constellation consists of a number of 6U nanosatellites in various orbits, none higher than 600 km. Early operational satellites, called OP1 constellation, occupy circular orbits at 500-700 km altitude and high inclination. Future launches will introduce satellites at lower inclinations to densify coverage of the mid-latitudes and tropics [16]. During the period of June-July 2019, the GNSS RO data are available from only two satellites (OP1-A and OP1-C). The CICERO satellites carry CION, which is a low-cost, low-power, and low-mass GNSS receiver developed at JPL [10]. The information regarding satellite orbit and the size and mass of the receiver is summarized in Table 1.

Table 1 Characteristics of currently active satellites in the CICERO constellation

Name	Orbit (perigee × apogee height / inclination angle)	Receiver (size / mass)
OP1-A (CICERO 7)	493 km × 504 km / 97.56°	CION (30 cm ×10 cm × 6 cm / 1.2 kg)
OP1-C (CICERO 8)	477 km × 499 km / 97.49°	

In this study, we used CICERO's Level 2 data '*cicPrf*', which contains the processed occultation profiles, including excess phase, bending angles, refractivity, and thermodynamic variables. We used the fourth version of the processed Level 2 data.

2.2. COSMIC I data

COSMIC I (FORMOSAT 3), which include six micro-satellites, is the first satellite constellation dedicated to remotely sensing Earth's atmosphere and ionosphere using GNSS RO [3]. The COSMIC I RO constellation carried the IGOR (Integrated GPS Occultation Receiver) GPS receiver, which is the improved version of the BlackJack receiver developed by JPL. Only one LEO satellite (C006) in the COSMIC I system was operational during June-July 2019, as COSMIC I was near the end of its lifespan.

Table 2 Characteristics of COSMIC I's 6th LEO satellite

Name	Orbit (perigee × apogee height / inclination angle)	Receiver (size / mass)
FM6 (C006)	773.8 km × 831.2 km / 72°	IGOR (20 cm × 24 cm × 10.5 cm / 4.6 kg)

We used '*atmPrf*', which is the Level 2 data of COSMIC I; it contains full resolution profiles of physical parameters such as dry pressure, dry temperature, refractivity, bending angle, impact parameter versus geometric height above mean sea level [17]. The CDDAC also provides radiosonde data, '*sonPrf*', which contains temperature, pressure, and refractivity data generated from radiosonde data collocated with occultation profiles. These radiosonde profiles are generally used for comparison with COSMIC I's RO profiles near the radiosonde. In the selected data set, the difference in latitude between the occultation point of COSMIC 1 and the point of radiosonde is up to 5° and the difference in longitude is up to 9°.

3.0. ASSESSMENT METHODOLOGY

In this study, we assessed the performance of CICERO by data analysis of several aspects, and then compared the results with that of COSMIC I. The data analysis was performed for the following three characteristics: geographic distribution, altitude distribution, and refractivity error.

As the quality of weather models is highly dependent on the spatial and temporal distributions of RO measurements and global data acquisition capability, the geographic distribution of RO occultation measurements is an important characteristic to be analyzed for RO performance evaluation. We compared the geographic histogram of the selected 12 days' occultation measurements from both CICERO and COSMIC I. The geographic histogram of the occultation measurements was obtained by counting the number of occultation measurements in each longitude-latitude grid.

The capability of the RO to obtain key weather parameters in the lower troposphere depends on its ability to penetrate low altitudes; therefore, altitudinal analysis is necessary. We investigated the penetration altitude and penetration percentage to analyze the altitude distribution. Penetration altitude is the minimum altitude of each RO sounding above the local surface [18], while penetration percentage is the percentage of RO events penetrating to a given altitude.

Because the refractivity is the RO's primary measurements from which atmospheric parameters can be derived, an analysis of refractivity error is essential. Collocated profiles obtained from CICERO, and COSMIC I within a defined time window were selected to compute the refractivity error of RO measurements with respect to radiosonde measurements in nearby areas. In this study, we defined the time window as one hour and the maximum difference for the collocation condition as 20° in both latitude and longitude. The refractivity errors, which are the refractivity differences between RO measurements (N_{RO}) and radiosonde observables ($N_{Radiosonde}$), are computed for both CICERO and COSMIC I and compared to each other.

4.0. PERFORMANCE ASSESSMENT RESULTS

4.1. Geographic distribution of RO measurements

We used the RO measurements obtained below 10 km altitude (troposphere), which contain important atmospheric parameters for numerical weather prediction. The total numbers of measurements (below 10 km altitude) from COSMIC I and CICERO are 406,539 and 384,836, respectively.

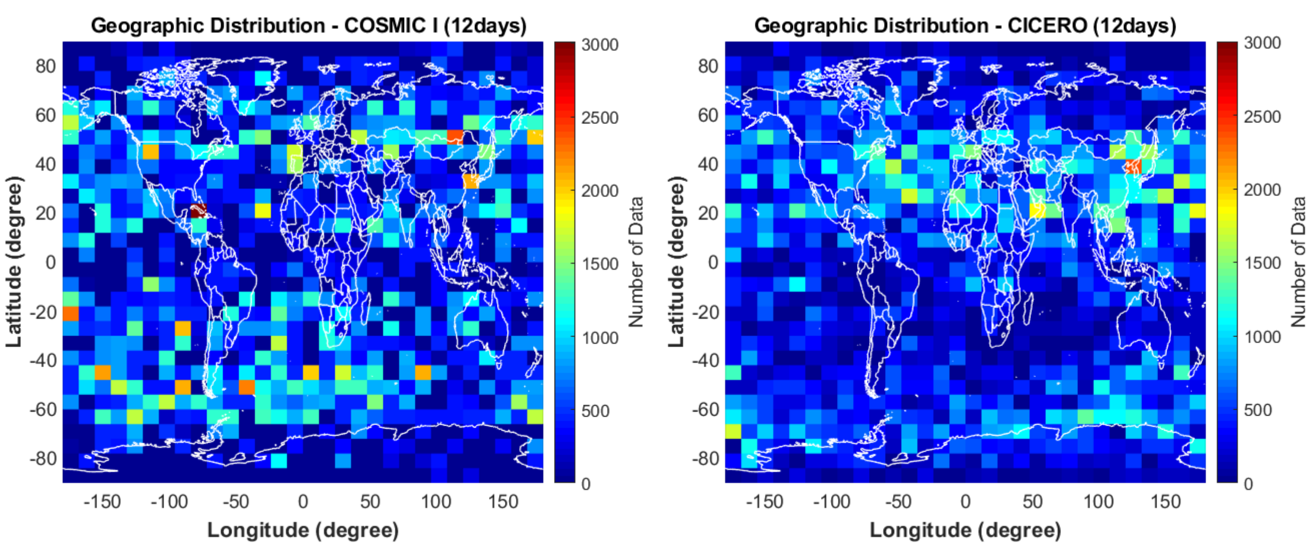

Figure 1 Geographic distribution of COSMIC I (left) and CICERO (right) RO measurements below 10 km

Figure 1 shows the geographic distributions of COSMIC I and CICERO RO measurements collected on the 12 selected days. The horizontal axis divides the longitude into bins. The vertical axis divides the latitude into bins, and the color scale of each pixel indicates the number of measurements within the bin (the color scale from blue to red shows an increase of the number of measurements). The RO measurements of both COSMIC I and CICERO show global coverage; however, their distributions are not uniform.

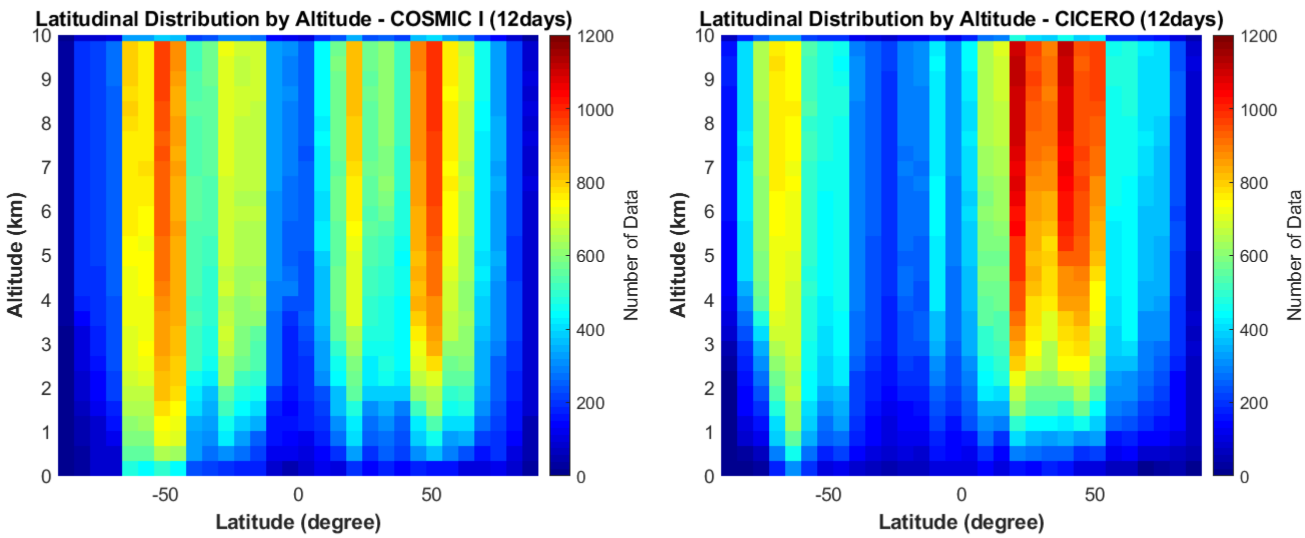

Figure 2 Latitudinal distribution of COSMIC I (left) and CICERO (right) RO measurements

Figure 2 presents the latitudinal distributions of COSMIC I (left) and CICERO (right) RO measurements below 10 km of altitude. Both COSMIC I and CICERO measurements have symmetric distributions with respect to the Equator; however, they are not uniformly distributed. This is because both COSMIC I and CICERO constellations have high inclinations (72° for COSMIC I and 97° for CICERO); hence, the RO measurements are more concentrated in mid-latitude regions. In addition, because the CICERO constellation includes near-polar orbiting LEO satellites, the latitudinal histogram in Fig. 3 shows that CICERO has more measurements in high latitude regions (red-shaded region having a range between -90° to -70° and 70° to 90° in latitude).

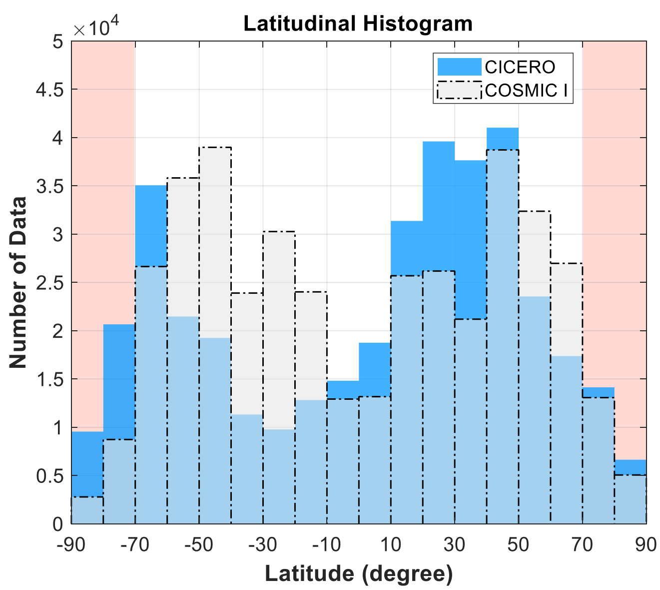

Figure 3 Latitudinal histogram of CICERO and COSMIC RO measurements

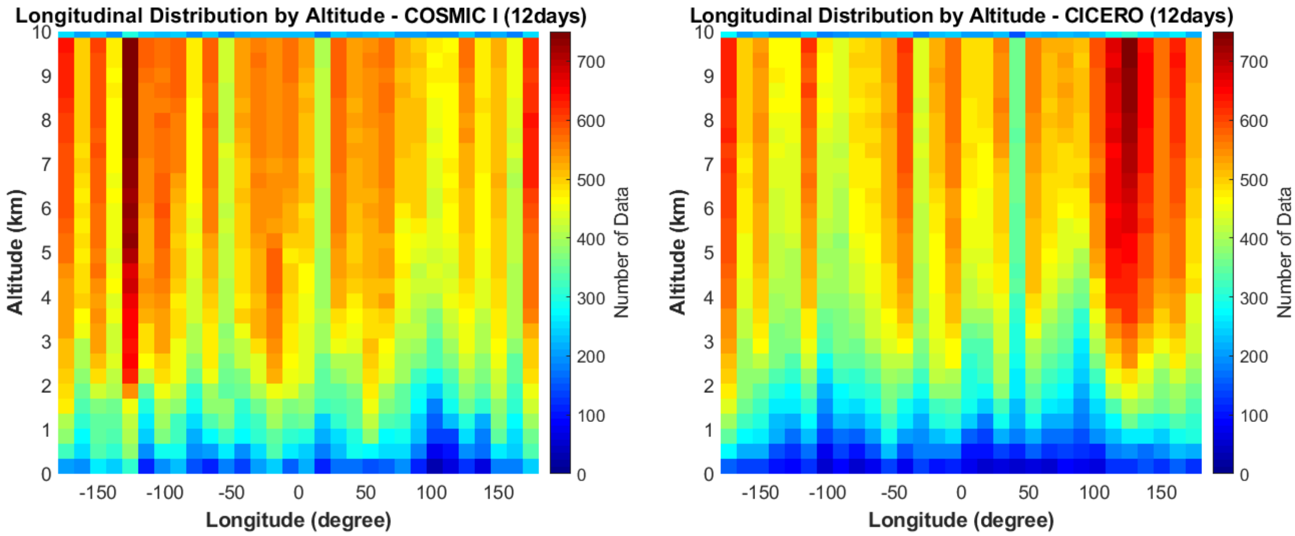

Figure 4 Longitudinal distribution of COSMIC I (left) and CICERO (right) RO measurements

Figure 4 shows the longitudinal distributions of COSMIC I (left) and CICERO (right) RO measurements below 10 km of altitude. The longitudinal distributions are more uniform than the latitudinal distributions. This is a clear result of longitudinal uniformity of GPS constellation geometry observed from the Earth's surface [19]. The longitudinal distribution of CICERO has a heavy concentration in the 100° ~ 180° longitude area, compared to that of COSMIC I.

4.2. Altitude (penetration) distribution of RO measurements

Figure 5 shows the global map of the penetration altitude for COSMIC I (left) and CICERO (right). The color scale bar represents penetration altitudes of RO measurements indicating a higher penetration ability as the altitude decreases. The regions with high penetration altitude (yellow to red color) are polar regions and mountainous regions (e.g., The Rocky mountains, Andes mountains, and Himalayas). Because the lowest altitude where RO profiles can reach is limited to the terrain elevation, the polar and high mountain regions (whose elevations are higher than other continents) have higher penetration altitude values. The percentage of the region which has a penetration altitude lower than 1 km is 61 % for COSMIC I and 47 % for CICERO. The results show that COSMIC I has a wider coverage of penetrate altitude below 1 km compared with CICERO.

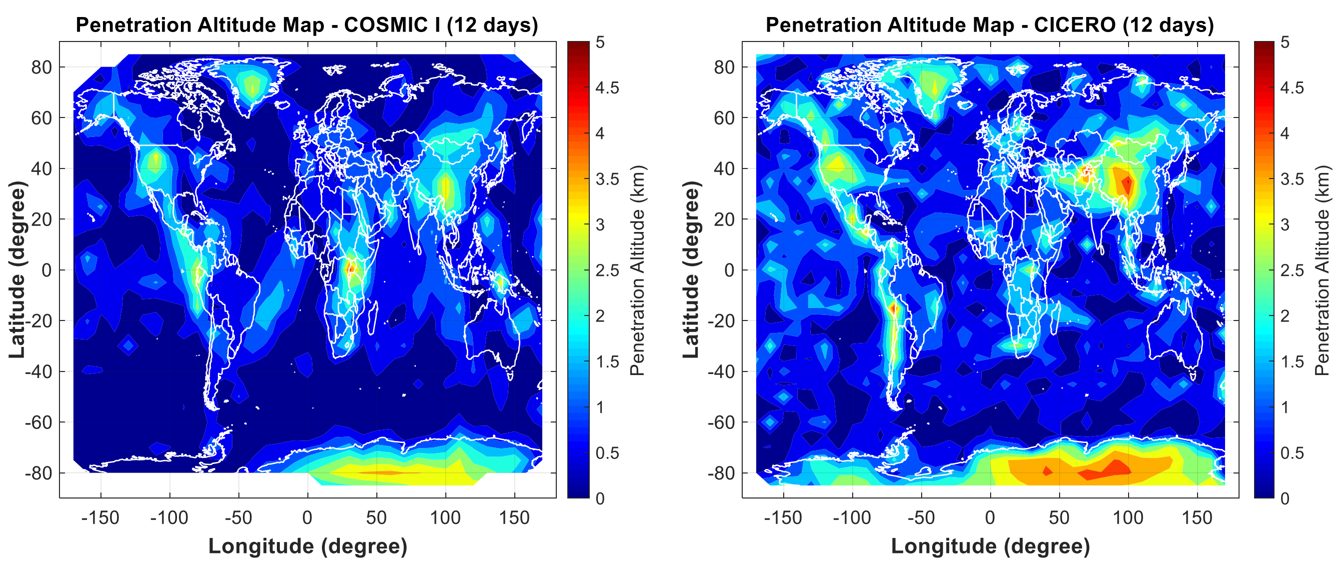

Figure 5. Penetration altitude map for COSMIC I (left) and CICERO (right) from total 12 days' data

Figure 6 shows the percentage of RO events penetrating different altitudes (as indicated by the horizontal axis) for COSMIC I (curves with black circles) and CICERO (curves with blue diamonds). At altitudes between 6 to10 km, the penetration percentage of CICERO is similar to those of COSMIC I. However, the difference between the two graphs is more apparent at lower altitudes. The percentages of RO events that penetrate below an altitude of 1 km are 70 % and 60 % for COSMIC I and CICERO, respectively. Although the results show that CICERO performs worse than COSMIC I, CICERO also has a high penetration capability compare to other RO systems, such as CHAMP and GPS/MET which have penetration percentages lower than 45 % (below 1 km) [20].

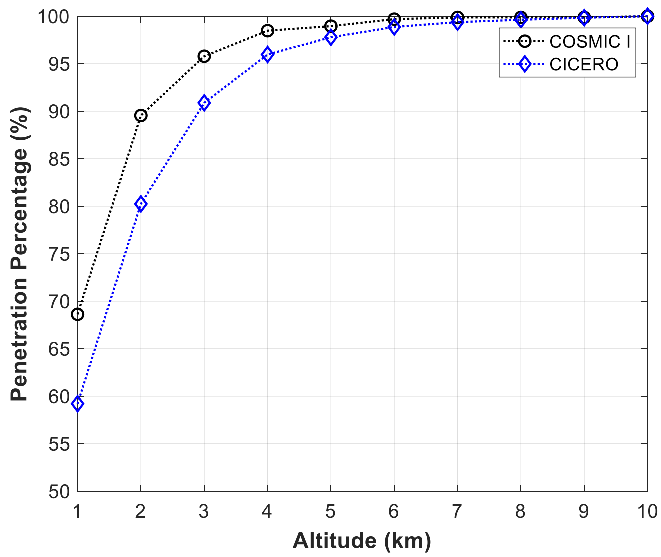

Figure 6. Penetration percentages for COSMIC I (circle) and CICERO (diamond)

4.3. Results of refractivity error analysis

As mentioned in Section 3, the collocation pairs were determined based on a spatial difference of less than 20° in both latitude and longitude and a time window of less than one hour. Figure 7 shows the results from one of the selected pairs (pair #1) on July 10, 2019. Occultation starting times were 17:59:00 for COSMIC I and 17:19:30 for CICERO. In Fig. 7, the left plot shows the locations of the COSMIC I occultation point (black circle), CICERO measurements (blue diamond), the CICERO occultation point (red-lined diamond) and radiosonde (green square). The middle plot shows the refractivity from COSMIC I (black-dot), CICERO (blue-dot with dashed line) during the period of 17:20:27 – 17:20:42, and radiosonde (green-square with sold line). The right plot shows the refractivity difference with respect to collocated radiosonde measurements for both COSMIC I ($N_{COSMIC} - N_{Radiosonde}$; plotted as gray circles) and CICERO ($N_{CICERO} - N_{Radiosonde}$; plotted as light-blue diamonds). The plot of $N_{CICERO} - N_{Radiosonde}$ is similar to that of $N_{COSMIC} - N_{Radiosonde}$ at upper altitudes between 8 to 10 km; however, it has a completely different shape from $N_{COSMIC} - N_{Radiosonde}$ plot at lower altitudes below 8 km, as shown in Fig. 7 (right). Figure 8 shows the same information in Fig. 7; however, the data is from collocation pair #2, which has a smaller spatial difference than that of pair #1. Occultation starting times were 17:59:00 (the same as pair #1) for COSMIC I and 17:18:12 for CICERO. The refractivity from CICERO during the period of 17:19:22 – 17:19:32 is shown in Fig. 8 (middle). The latitude difference between the COSMIC occultation point and the CICERO occultation point was -3.6814°, and the longitude difference was -4.5146°. When compared with the plots of refractivity difference in Fig. 7 (right), the plots of $N_{CICERO} - N_{Radiosonde}$ and $N_{COSMIC} - N_{Radiosonde}$ in Fig. 8 (right) have nearly identical shapes.

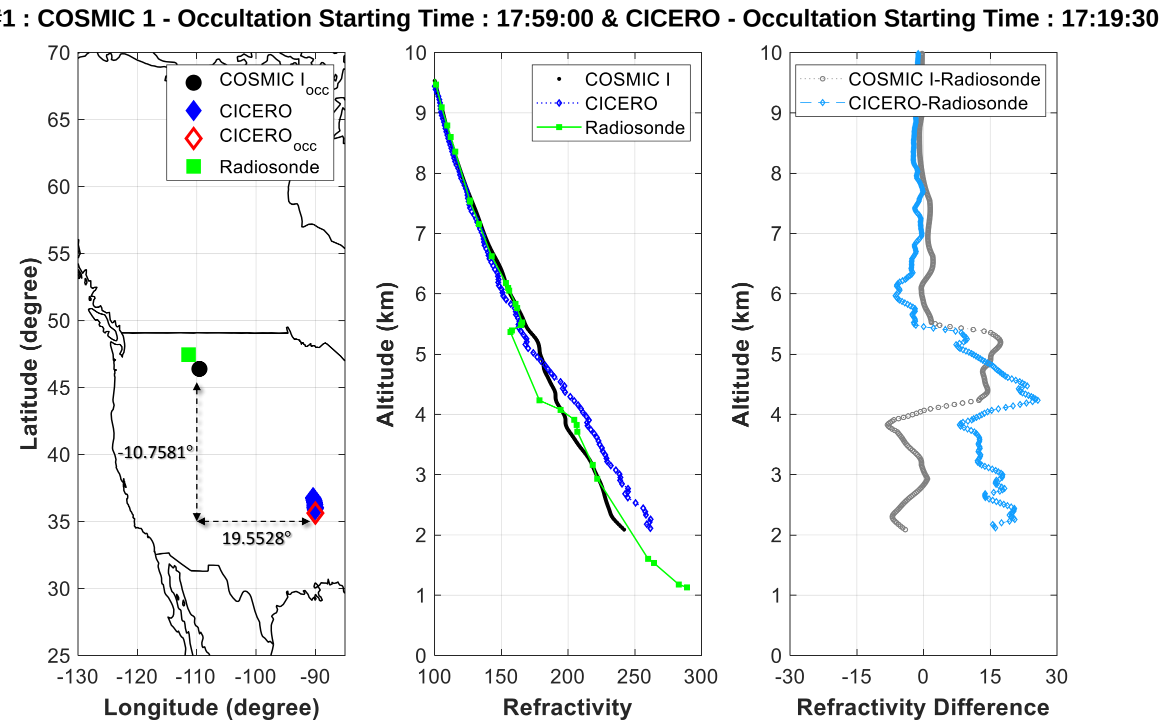

Figure 7. Locations (left), refractivity (middle) and refractivity difference (right) of Pair #1 (starting time of COSMIC I occultation : 17:59:00 & starting time of CICERO occultation : 17:19:30)

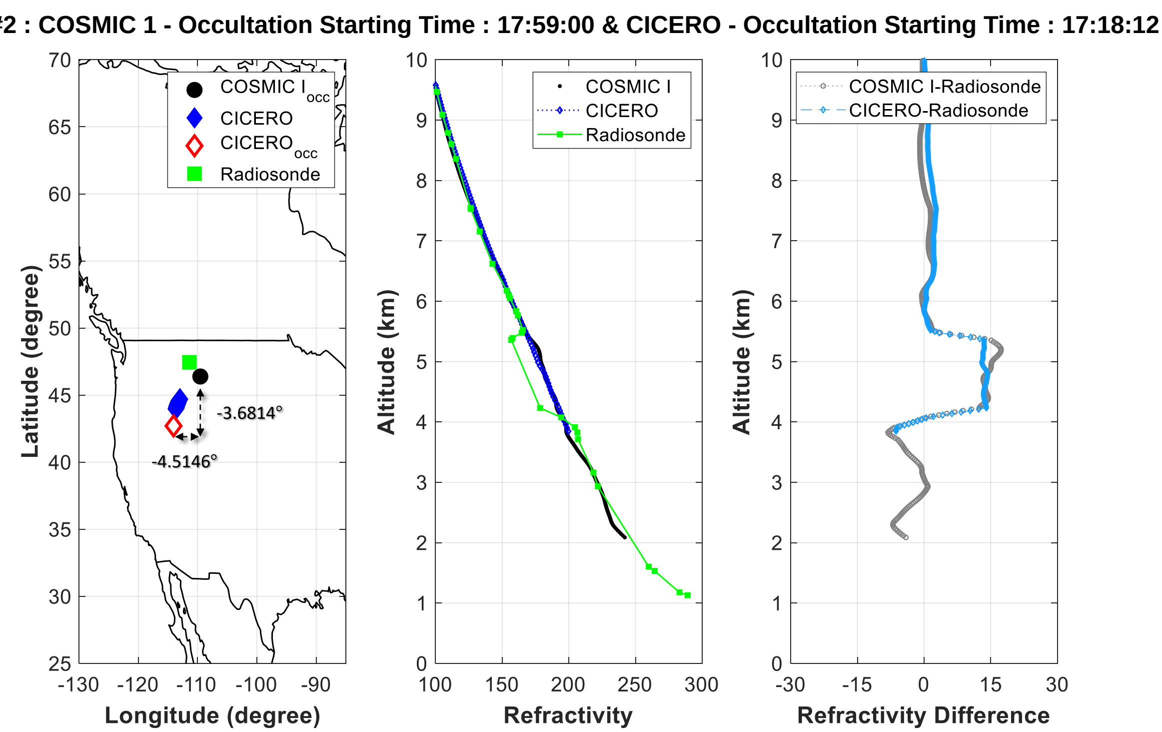

Figure 8. Location of (left), refractivity (middle) and refractivity difference (right) of Pair #2 (starting time of COSMIC I occultation : 17:59:00 & starting time of CICERO occultation : 17:18:12)

Based on the results from Fig. 7 and 8, it is evident that spatial and temporal mismatches could increase the estimates of refractivity errors, and the effect of spatial mismatches is larger than that of temporal mismatches. Thus, we also analyzed the impact of spatial mismatches between RO occultation points and the location of radiosonde on refractivity comparisons. The mismatches are defined with latitude and longitude differences, as shown in Fig.9. A total of five cases were examined with the criteria of latitude and longitude differences (Table 3). The latitude and longitude differences in case number 4 correspond to the above-mentioned maximum spatial difference between COSMIC I occultation points and the locations of collocated radiosonde.

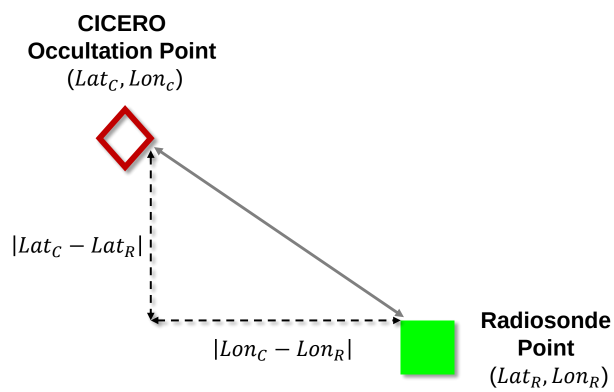

Figure 9. Spatial mismatches (spatial difference) configuration

Table 3. Spatial difference criteria for each case

Case Number (#)	Maximum value of $|Lat_C - Lat_R|$	Maximum value of $|Lon_C - Lon_R|$
1	20°	20°
2	15°	15°
3	10°	10°
4	5°	9°
5	5°	5°

Figure 10 shows the difference between RO and radiosonde refractivity measurements for each case. In each plot, the refractivity difference between CICERO and radiosonde data are plotted as asterisks, and the difference between COSMIC I and radiosonde refractivity are plotted as black circles. As the criteria for spatial mismatches become more stringent (i.e., smaller size of the spatial difference), the refractivity difference between CICERO and radiosonde data tends to follow the refractivity difference between COSMIC I and radiosonde data.

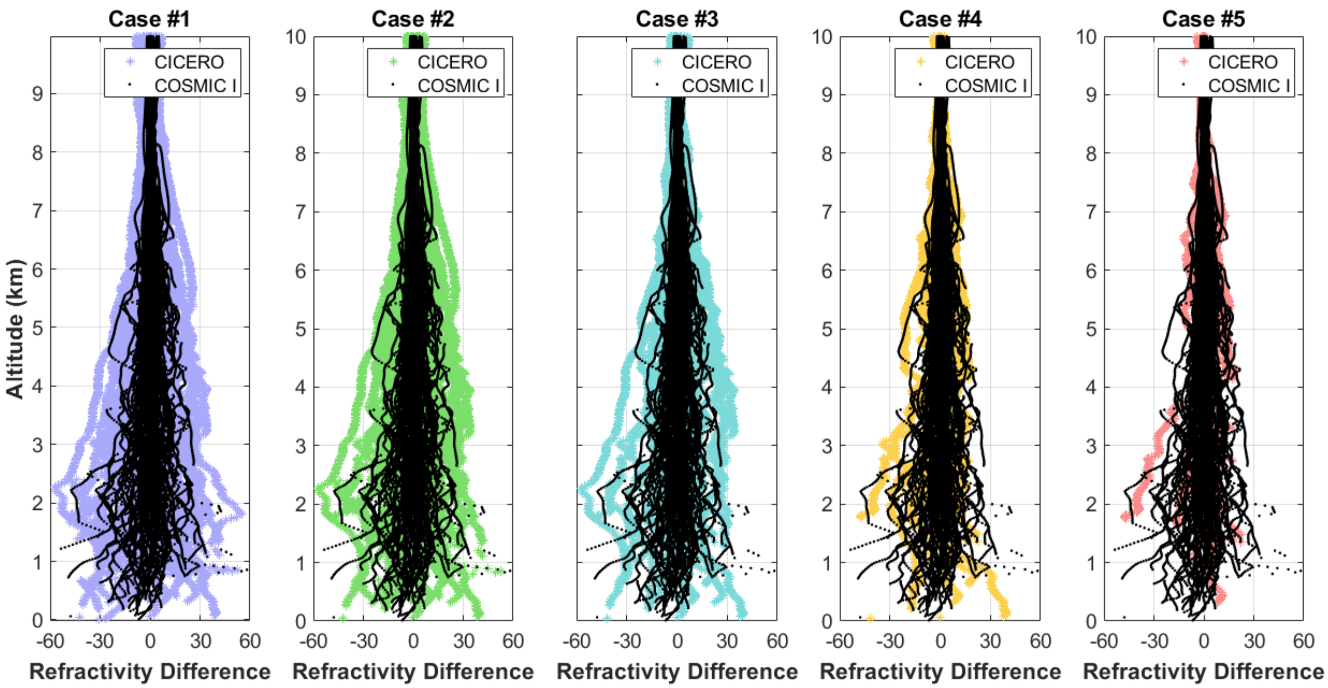

Figure 10. Refractivity difference between RO and radiosonde data for each case

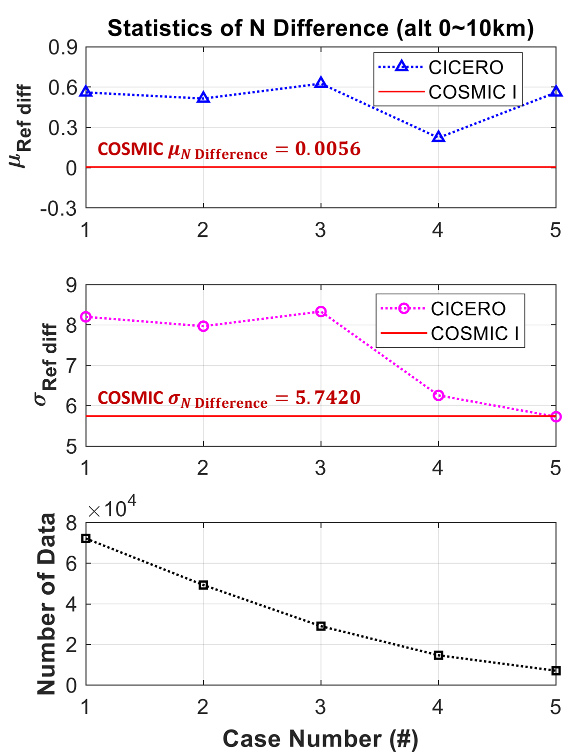

Figure 11. Statistics of refractivity difference

In addition, the statistics of refractivity error estimates were also analyzed for each case. The top plot in Fig. 11 shows the mean refractivity differences with respect to radiosonde data for CICERO (blue triangle) and COSMIC I (red line) for each case. The standard deviations of refractivity differences for CICERO (pink circle) and COSMIC I (red line) are shown in the middle plot, and the number of data for each case is presented in the bottom plot. The mean refractivity difference of COSMIC I is 0.0056 and the standard deviation (1-sigma) is 5.7420. The mean and sigma values of the refractivity difference for CICERO are similar to those of COSMIC I for case number 4, because the criterion for case number 4 is equal to the maximum spatial difference between the COSMIC I occultation point and radiosonde location.

We also evaluated the impact of refractivity errors on the estimation of temperature, which is one of the major atmospheric parameters, by using the fractional refractivity difference. The fractional refractivity difference is the parameter used in several previous studies for analysis of radio occultation profiles [18, 21-22]. The fractional refractivity differences are calculated with the following equation (1);

$$\text{Fractional Refractivity Difference (\%)} = \frac{N_{RO} - N_{radiosonde}}{N_{radiosonde}} \times 100 \tag{1}$$

The following equation for the refractivity (N) was used to quantify the impact of the fractional refractivity differences on differences in temperature [23].

$$\text{Refractivity } (N) = 77.6\frac{P}{T} + 3.73 \times 10^5 \frac{P_W}{T^2} \tag{2}$$

T is temperature in K, P and P_W are the total air pressure and partial pressure of water vapor in hPa, respectively.

Only the dry component of the temperature in the upper troposphere, where the temperature T is lower than 250 K, was analyzed in this study to assess the performance without any assumptions regarding the retrieval process. This is because, at altitudes where water vapor is considerable (approximately below 5 km in altitude), the retrieval of the atmospheric parameters is significantly dependent on external information and assumptions [24]. At dry atmosphere altitudes, where the water vapor pressure is low, the P_W in equation (2) could be negligible and thus, a percentage difference in refractivity corresponds to the same percentage difference in temperature. For example, for a temperature of 250K, a 1 % difference in N corresponds to a 2.5 K difference in temperature [21]. The range of dry atmosphere altitude for this analysis is between 5 and 15 km. With the same spatial mismatches of case number 4, the mean value of fractional refractivity differences for CICERO is -0.1 %. This value could correspond to a temperature difference of approximately 0.2 K. A previous study presented that the required accuracy of temperature for stringent climate monitoring is 0.5 K [25]. This magnitude is more than twice that of the temperature difference magnitude from the CICERO data. Therefore, we conclude that this degree of the estimation error is acceptable and comparable to the conventional COSMIC I value.

5.0. CONCLUSIONS AND FUTURE WORK

In this study, we evaluated the performance of newly developed CICERO in several aspects by comparing it with that of existing radio occultation of COSMIC I. This study shows that CICERO can acquire global data coverage at low altitudes as well as COSMIC I. The results show that the refractivity error of CICERO is similar to that of COSMIC I when the spatial mismatches of CICERO are similar to the differences between COSMIC I occultation point and radiosonde location. Furthermore, the analysis results of the refractivity error impact on temperature demonstrates that the miniature version of the GNSS RO receiver in CICERO can satisfy certain accuracy requirements of climate monitoring. The results of this study demonstrate that low cost, low volume, and low power LEO cubeSats can potentially improve data obtainability and accuracy for weather prediction.

The subsequent version of the COSMIC I—COSMIC 2 (also known as FORMOSAT 7) —was launched on June 25, 2019. COSMIC 2 carries the TriG RO receiver, which offers enhanced performance compared to COSMIC I. The performance assessment of CICERO with respect to the recent COSMIC 2 can be further conducted with newly provided COSMIC 2 measurements from CDDAC.

ACKNOWLEDGMENTS

The CICERO data used in this study is provided by GeoOptics under an agreement with the University of Colorado. The COSMIC I data is obtained from NASA CDDAC website. This research was supported by the MSIT (Ministry of Science, ICT), Korea, under the High-Potential Individuals Global Training Program (No. 2019-0-01598) supervised by the IITP (Institute for Information & Communications Technology Planning & Evaluation).

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